Enterprise RAG Assistant

Software Requirements Specification

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1 Introduction

1.1 Purpose

This SRS defines the required behaviour, interfaces, data rules, and quality targets of Enterprise RAG Assistant. It gives the project owner, developer, tester, and course reviewer a common basis for implementation and acceptance. It describes the completed intended first release, including functions still under development.

1.2 Scope

The assistant will answer employee questions in English using approved company documents. Each substantive answer will cite permitted sources; insufficient evidence will produce a generic not-found result. Version 1 includes local Markdown and text-based PDF ingestion, authentication, department access, a REST API, browser chat, query metrics, evaluation, and a deployed demonstration.

The demo represents one fictional company, Acme Corp, using public GitLab Handbook pages. Voice, multilingual support, native mobile apps, SSO, model training, document editing through chat, live synchronisation, scanned-PDF OCR, and multi-tenant hosting are outside this release.

1.3 Definitions and Conventions

RAG means retrieval-augmented generation: generating an answer from retrieved passages. A chunk is a passage linked to its source document. RBAC means role-based access control. A citation identifies the source title and section or PDF page. Shall denotes a requirement. FR and NFR IDs retain the project baseline; other prefixes identify supporting requirement categories.

1.4 References

The baseline is the project Requirements Document v0.2, Design Document v0.3, and data-source records. The supplied SRS.pdf and annotated SRS template guide the structure. Full source details and dataset attribution appear in Section 6.3.

1.5 Document Overview

Section 2 describes context, users, scope limits, and dependencies. Section 3 specifies the eight required requirement categories within the standard interface, function, performance, data, constraint, and quality sections. Sections 4 to 6 cover changes, approval, verification, and references.

2 Overall Description

2.1 Product Perspective and Operating Context

The assistant serves one organisation’s document collection. It supplies information rather than carrying out payroll, leave, purchasing, or approval transactions.

2.1.1 System interfaces Inputs are credentials, questions, approved files, and evaluation records. Outputs are answers, citations, metrics, and reports.

2.1.2 User interfaces Employees use browser login and chat screens; the operator uses command-line ingestion and evaluation tools.

2.1.3 Hardware interfaces No specialised hardware interfaces are required.

2.1.4 Software interfaces A hosted model service receives questions and permitted evidence. Internal components follow the design document.

2.1.5 Communications The deployed client and API use HTTPS with bearer tokens; generation requires outbound provider access.

2.1.6 Memory constraints No RAM, storage, or concurrency limit is approved; these require baseline measurement.

2.1.7 Operations Ingestion runs offline while serving is stopped. Restart checks index consistency. Evaluation is an operator activity.

2.1.8 Site adaptation The operator configures sources, roles, credentials, storage, retrieval, and logging.

2.2 Product Functions

The system ingests and labels documents, authenticates employees, retrieves permitted evidence, generates answers, validates citations, records metrics, and evaluates results.

2.3 User Characteristics

Employees need English reading skills and ordinary browser experience. Operator is a maintenance responsibility, not a fifth employee role.

Role	Permitted departments
New Hire (new_hire)	onboarding, general
HR (hr)	people, onboarding, general
Finance (finance)	finance, onboarding, general
Engineer (engineer)	engineering, onboarding, general

2.4 Constraints

Version 1 uses one API worker, offline ingestion, department-level permissions, and one organisation. Backend checks enforce access before generation.

2.5 Assumptions and Dependencies

The operator approves sources and classification. Network and model-provider access are required. No enterprise stakeholder interview has been completed; quality targets remain provisional.

2.6 Apportioning of Requirements

SSO, immediate revocation, background synchronisation, finer permissions, and multiple workers require a later scope change.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User interfaces The browser shall provide credential and question entry, answers, citations, and request feedback. Employees shall not assign their own role or upload approved documents.

3.1.2 Hardware interfaces No direct device interfaces apply.

3.1.3 Software interfaces The API shall expose the operations below. The model service shall receive only the question and permitted evidence.

Operation	Input and result
POST /auth/login	Credentials; signed token on successful authentication.
POST /ask	Question and verified bearer token; answer, citations, found flag.
GET /health	No token in the current design; service and index-consistency status.

3.1.4 Communications Deployed requests shall use HTTPS. Exact JSON fields, input limits, error codes, provider settings, and retry bounds remain to be agreed.

3.2 Functions and Related Requirements

3.2.1 Business Requirements

BREQ-01 The solution shall provide one place to ask questions about approved documents, reducing separate manual searches.

BREQ-02 The solution shall make answers traceable to their sources.

BREQ-03 The solution shall preserve the department-access policy.

BREQ-04 The project shall provide repeatable evidence of quality, safety, response time, and cost before acceptance.

3.2.2 User Requirements

UR-01 Employees shall be able to sign in and ask questions in ordinary English through a browser.

UR-02 Employees shall be able to read an answer with its document title and section or page reference.

UR-03 Employees shall receive information only from departments permitted for their authenticated role.

UR-04 Employees shall receive not found without learning whether restricted evidence exists.

UR-05 The operator shall be able to prepare documents and run evaluation through command-line tools.

3.2.3 Domain Requirements

DOM-01 Sources shall belong to onboarding, general, people, finance, or engineering. Unknown roles and departments shall receive no default permission.

DOM-02 Classification shall use the longest approved path prefix; file mappings override folders. finance/readmes and finance/jihu-support shall be excluded.

DOM-03 The pinned demo shall contain 190 pages: engineering 60, finance 35, people 50, onboarding 5, general 40; each shall meet the documented 150-real-word minimum.

DOM-04 Dataset records shall retain provenance and SHA-256 checksums with CRLF normalised to LF. Role restrictions are simulated over public handbook material.

3.2.4 Functional Requirements

All functional requirements are required for version 1. Access checks precede generation, and every substantive answer requires a valid source.

FR-01 The system shall ingest Markdown and text-based PDF documents from a configured local folder.

FR-02 The system shall attach department and source metadata to each ingested document and its chunks, preserving the information needed for access checks and citations.

FR-03 The system shall answer a natural-language question using retrieved document content as its factual evidence.

FR-04 The system shall include at least one citation containing a document title and section or page in every substantive answer.

FR-05 The system shall retrieve and supply evidence only from departments permitted for the authenticated employee's role.

FR-06 The system shall return not found when the permitted evidence cannot support an answer.

FR-07 The system shall provide question answering through a REST API and a browser-based chat interface.

FR-08 The system shall log role, retrieved chunk IDs, relevance scores, latency, and token count. Question and answer text shall be stored only when content logging is enabled.

FR-09 The system shall provide an evaluation script that runs the defined question set and reports per-question and aggregate results.

FR-10 The system shall preserve its access and answering rules when a question or document contains instructions that attempt to override them.

FR-11 The system shall authenticate employees and determine their role from a signed token verified by the backend.

FR-12 The system shall return the same generic not-found response for restricted evidence and evidence that does not exist.

FR-13 The system shall discard citation IDs not supplied to the model and return not found if no valid citation remains. It shall not make another model call solely to repair citations.

3.2.5 Inverse Requirements

These requirements state prohibited behaviour; their supporting functions remain specified above.

INV-01 The system shall not invent an answer when permitted evidence is insufficient. [FR-03, FR-06]

INV-02 The system shall not disclose restricted passages, citations, or document titles. [FR-05, FR-12]

INV-03 The system shall not trust a client-supplied role or follow instructions that bypass access rules. [FR-10, FR-11]

INV-04 The system shall not release an answer with no valid citation. [FR-04, FR-13]

INV-05 The system shall not ingest excluded, unmapped, or malformed sources under a fallback department. [DOM-01, DOM-02]

INV-06 The system shall not log question text when content logging is disabled, commit secrets, edit policies, or perform their business transactions. [FR-08, NFR-07; Section 2.1]

3.2.6 System Requirements

SYS-01 The system shall retain source identity, title, department, checksum, chunk text, and citation locators. Chunk IDs shall be unique.

SYS-02 Both vector and keyword retrieval shall restrict evidence to permitted departments. Keyword filtering shall occur before final top-k selection.

SYS-03 Evidence sufficiency shall use reranker relevance when enabled, otherwise best vector similarity. A fused RRF score shall not determine refusal; thresholds shall be calibrated per configuration.

SYS-04 The chunk collection shall be authoritative for both indexes. Offline updates shall replace changed-document chunks, remove deleted sources, and avoid re-embedding unchanged documents.

SYS-05 Startup shall compare chunk counts and checksums across the collection and indexes and refuse to serve when they disagree.

SYS-06 The backend shall verify question-request tokens and derive roles from them. Tokens expire after 30 minutes in the current design; immediate revocation is deferred.

SYS-07 Question responses shall contain answer, citations, and a found flag. Provider failure shall be a service error, distinct from not found.

SYS-08 Metrics shall be written asynchronously. Client deployments shall disable content logging by default; logged role values shall be bounded and sanitised.

3.3 Performance and Non-Functional Requirements

These numerical targets are provisional and retain the project baseline. Test conditions require approval before a pass is declared. The other non-functional attributes appear in Section 3.6.

NFR-01 Response time At least 95% of valid answer requests shall complete within 8 seconds, measured from API receipt to the complete response under the agreed workload.

NFR-02 Answer quality Faithfulness shall be at least 0.85 and answer correctness at least 75% on the agreed question set. The scoring rubric and refusal treatment require approval.

NFR-04 Cost Average model cost shall not exceed USD 0.02 per evaluated question. The report shall record tokens, price basis, question count, and accounting scope.

3.4 Logical Database Requirements

A document has document_id, source_path, title, department, and checksum. Its chunks have unique chunk_id, document_id, department, text, and section or page locator. Query records contain role, chunk IDs, scores, latency, and tokens; text is optional only under the logging switch.

The manifest contains source_path, department, and sha256; paths shall be unique and sorted. Evaluation questions contain id, role, question, expected_answer, source_doc, and should_refuse. Retention, deletion, credentials storage, and backup policy remain open.

3.5 Design and Implementation Constraints

DC-01 The first release shall use one API worker. Ingestion shall run offline while serving is stopped, followed by restart and consistency checking.

DC-02 The backend shall enforce permissions before generation. UI controls and prompts shall not be the access-security boundary.

DC-03 Ingestion and evaluation shall remain operator command-line tools, with no employee API for modifying the document collection.

DC-04 The demonstration shall retain the pinned handbook revision and its licence attribution.

DC-05 The design shall maintain one role-policy table and isolate the model provider and vector store behind replaceable interfaces.

The planned stack is Python 3.12, FastAPI, Streamlit, SQLite, ChromaDB, BM25, local embeddings, a reranker, and Groq. These are project design choices, not customer-mandated technologies. No specialised hardware or fixed memory ceiling has been approved.

3.5.1 Standards Compliance

The supplied course handout and SRS template guide this document. The handbook attribution record identifies CC BY-SA 4.0. No independently audited standards, regulatory, accessibility, or security certification is claimed.

3.6 Software System Attributes

3.6.1 Reliability Unsupported evidence shall produce not found; invalid citations shall be removed; inconsistent indexes shall block serving. No numerical MTBF target is approved. Provider failure is a separate service error.

3.6.2 Availability Updates require downtime. An uptime percentage, recovery-time target, and recovery-point target remain to be agreed.

3.6.3 Security and privacy Backend token verification and department filtering govern access. Automatic personal-data masking and immediate token revocation are deferred.

NFR-03 Access safety No restricted content shall be disclosed across the defined access tests. Answers, citations, retrieved chunks, and model inputs shall be inspected; any leak fails this criterion.

NFR-07 Secret protection Keys, passwords, signing secrets, and other deployment secrets shall not be committed to the repository; tracked files and exclusion rules shall be inspected.

3.6.4 Maintainability Component changes shall preserve access, refusal, and citation behaviour and pass the affected verification procedures. No numerical complexity target is approved.

3.6.5 Portability Development uses Windows and PowerShell; project tests are intended to remain compatible with Linux. The supported host and browser matrix remains open.

NFR-05 File compatibility Markdown and text-based PDFs shall be supported; scanned-PDF OCR is excluded from acceptance.

NFR-06 Reproducibility Setup from the README using Docker shall take less than 30 minutes on a clean supported host under agreed conditions, with prerequisites and downloads recorded.

Usability and accessibility: the interface shall provide login, question entry, readable answers, citations, and request feedback. Formal accessibility conformance and measured usability targets require agreement.

4 Change Management Process

The owner shall record proposed changes with affected IDs, revised wording, rationale, and effects on scope, access, data, implementation, and tests. Accepted changes update the SRS version, design references, and verification records. Withdrawn IDs are not reused for unrelated obligations. Changes to an academically approved baseline are submitted to the reviewer when required.

5 Document Approvals

Prepared by Abdul Samad. Project-owner and course-instructor approval are pending; no signature, approval date, or completed acceptance result is implied. Document approval establishes the requirements baseline; product acceptance follows implementation and testing.

6 Supporting Information

6.1 Verification and Traceability

The core acceptance set has 30 planned source-based questions: 12 factual, 6 multi-section/table, 6 restricted-access, and 6 outside the documents. Additional fixtures cover authentication, injection, citations, source updates, and index integrity. Permission tests cover every role and allowed/denied department pair.

Requirement group	Verification and source
BREQ and UR	Demonstrate login, question answering, source lookup, and role outcomes; project requirements [1].
FR-01 to FR-13	Test ingestion, answers, citations, refusal, API/chat, logs, evaluation, injection, and token handling; [1], [2].
NFR-01 to NFR-07	Measure timing, quality, cost, and setup; inspect access safety, file support, and secret exclusions; [1].
DOM, INV, SYS, DC	Check mappings, manifest, prohibited actions, model inputs, filtering order, updates, startup consistency, and configuration; [2], [3].

6.2 Open Decisions

Before acceptance, agree the host and concurrency; scoring method and thresholds; cost boundary; retention and recovery; credential and provider-data handling; input/error contracts; and hosting, browser, and accessibility scope. Provisional targets are not measured achievements. A leak fails access safety; unexecuted or failed checks remain visible.

6.3 References and Attribution

[1] Abdul Samad, Requirements Document: Enterprise RAG Assistant, v0.2, docs/01-requirements.md. [2] Design Document, v0.3, docs/02-design.md. [3] docs/data-sources.md and docs/data-manifest.csv.

[4] Supplied SRS.pdf and Software Requirements Specification SRS Template. The template’s alternative organisations are guidance, not additional product functions.

[5] GitLab Handbook, https://handbook.gitlab.com, recorded under CC BY-SA 4.0: https://creativecommons.org/licenses/by-sa/4.0/. Pinned revision: 2c77a1f5b8c8a80cb7b5151ff11cfad84f98bfbb; download recorded as 11 October 2026.

Version 2.1 restores the standard SRS structure while retaining the eight required categories in seven pages. No handbook policy text is reproduced.